Boss Challenge — Paper 45 (Class 7)

Nutrition in Plants | Wastewater Story | Perimeter & Area | Exponents & Powers

One correct option each. Marking: +4 correct, -1 wrong, 0 blank. Write A/B/C/D on a separate sheet.

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1. A plant that builds its own food from simple raw materials, using sunlight, is said to follow the mode of nutrition called:

- **(A)** parasitic
- **(B)** autotrophic
- **(C)** saprotrophic
- **(D)** heterotrophic

2. The used, dirty water that drains from homes, factories and farms - water rich in waste - is given the general name:

- **(A)** distilled water
- **(B)** ground water
- **(C)** rainwater
- **(D)** sewage

3. The total distance once around the boundary of a closed flat shape is called its:

- **(A)** area
- **(B)** volume
- **(C)** diameter
- **(D)** perimeter

4. In the expression 5 raised to the power 3, the small raised number 3 is called the:

- **(A)** exponent
- **(B)** factor
- **(C)** product
- **(D)** base

5. The green colouring matter in a leaf that captures the energy of sunlight for food-making is named:

- **(A)** haemoglobin
- **(B)** cytoplasm
- **(C)** starch
- **(D)** chlorophyll

6. The network of underground pipes that carries sewage away from buildings to a treatment plant is the:

- **(A)** sewer system
- **(B)** water mains
- **(C)** canal
- **(D)** aqueduct

7. The amount of flat surface a shape covers, measured in square units, is its:

(A) height

(B) radius

(C) area

(D) perimeter

8. The value of 2 raised to the power 4 (that is 2 x 2 x 2 x 2) is:

(A) 16

(B) 8

(C) 12

(D) 6

9. The tiny pores on the surface of a leaf, through which gases move in and out, are called:

(A) stomata

(B) veins

(C) roots

(D) nodes

10. At a treatment plant the sewage is first passed through bar screens, whose job is to remove:

(A) large floating objects like rags and sticks

(B) dissolved salts

(C) harmful bacteria

(D) bad smell

11. For a rectangle whose length is l and whose breadth is b, the correct rule for its boundary length is:

(A) l times b

(B) 2 times (l + b)

(C) l + b

(D) 4 times l

12. The value of 10 raised to the power 4 is:

(A) 100000

(B) 1000

(C) 40

(D) 10000

13. Besides sunlight and chlorophyll, the two raw materials a green leaf needs to carry out photosynthesis are:

(A) nitrogen and salt

(B) oxygen and glucose

(C) carbon dioxide and water

(D) starch and water

14. After screening, the sewage flows slowly through a grit-and-sand removal tank so that:

- **(A)** chlorine kills germs
- **(B)** sand, grit and pebbles settle out
- **(C)** oil rises and is skimmed
- **(D)** bacteria are added

15. The area of a rectangle with length 9 cm and breadth 4 cm is:

- **(A)** 36 cm
- **(B)** 26 square cm
- **(C)** 36 square cm
- **(D)** 13 square cm

16. Written with a single exponent, the product 3 x 3 x 3 x 3 x 3 is:

- **(A)** 3 to the power 5
- **(B)** 3 to the power 4
- **(C)** 5 to the power 3
- **(D)** 15

17. The gas that a green plant releases into the air as a by-product of photosynthesis during the day is:

- **(A)** water vapour
- **(B)** nitrogen
- **(C)** carbon dioxide
- **(D)** oxygen

18. In a settling tank the cleared water moves on while the solids that sink to the bottom are collected as:

- **(A)** sludge
- **(B)** grit
- **(C)** scum
- **(D)** clarified water

19. A square has each side 6 cm long. Its perimeter is:

- **(A)** 36 cm
- **(B)** 24 cm
- **(C)** 12 cm
- **(D)** 10 cm

20. When multiplying powers of the same base, like 2^3 x 2■, the exponents are:

- **(A)** added, giving 2 to the power 7
- **(B)** multiplied, giving 2 to the power 12
- **(C)** kept the same, giving 2 to the power 4
- **(D)** subtracted, giving 2 to the power 1

21. When a few drops of iodine solution are placed on a leaf that has made food, the colour that appears is:

- **(A)** grass green
- **(B)** colourless
- **(C)** bright red
- **(D)** blue-black

22. Floatable oil, grease and fat that rise to the surface of the settling tank are removed by:

- **(A)** letting them settle at the bottom
- **(B)** filtering through sand
- **(C)** skimming them off the top
- **(D)** boiling the whole tank

23. A square garden plot measures 7 m along every edge. The surface it covers works out to:

- **(A)** 49 metres
- **(B)** 14 square metres
- **(C)** 28 square metres
- **(D)** 49 square metres

24. When dividing powers of the same base, like 5■ / 5■ (5 to the 6 divided by 5 to the 2), the exponents are:

- **(A)** multiplied, giving 5 to the power 12
- **(B)** subtracted, giving 5 to the power 4
- **(C)** added, giving 5 to the power 8
- **(D)** divided, giving 5 to the power 3

25. The amarbel (Cuscuta), a yellow climbing plant that twines on a host and draws food from it, is an example of a:

- **(A)** host
- **(B)** parasite
- **(C)** saprotroph
- **(D)** autotroph

26. After settling, the water is sent to an aeration tank where air is bubbled through it. This air feeds helpful microbes that:

- **(A)** colour the water blue
- **(B)** turn the water into drinking water at once
- **(C)** consume the dissolved and suspended organic waste
- **(D)** add minerals back to the water

27. A rectangular field is 40 m long and 25 m wide. The length of wire needed to fence it all the way round is:

- **(A)** 130 metres
- **(B)** 65 metres
- **(C)** 1000 metres
- **(D)** 80 metres

28. Following the laws of exponents, whenever a non-zero base is given the power zero, the result always comes out to be:

(A) 0
(B) the base itself
(C) undefined
(D) 1

29. An insect-trapping plant such as the pitcher plant catches insects mainly to obtain the nutrient it lacks in marshy soil, namely:

(A) carbon
(B) nitrogen
(C) water
(D) oxygen

30. The clarified water leaving the treatment plant, now much cleaner, is usually:

(A) poured back into the toilet
(B) burned as fuel
(C) sealed in tanks forever
(D) released into a river or sea, or used for gardening

31. The area of a right triangle is half the base times the height. A triangle of base 10 cm and height 6 cm has area:

(A) 16 square cm
(B) 30 cm
(C) 60 square cm
(D) 30 square cm

32. The number 64 can be written as a power of 2 because 2 x 2 x 2 x 2 x 2 x 2 equals 64. So 64 is:

(A) 6 to the power 2
(B) 2 to the power 6
(C) 2 to the power 7
(D) 2 to the power 5

33. A mushroom growing on a rotting log feeds by breaking down the dead wood for nutrients. Its mode of nutrition is:

(A) parasitic
(B) saprotrophic
(C) insectivorous
(D) autotrophic

34. Throwing cooking oil, paint and chemicals down the drain is harmful mainly because they:

(A) make the water taste sweet
(B) kill the helpful microbes that clean sewage
(C) turn sewage into drinking water
(D) help the pipes flow faster

35. A square and a rectangle have the same perimeter. The square's side is 8 cm; the rectangle is 10 cm long. The rectangle's breadth is:

(A) 8 cm
(B) 16 cm
(C) 4 cm
(D) 6 cm

36. In the number 7■, the base is:

(A) 5
(B) 35
(C) 12
(D) 7

37. Rhizobium bacteria living in the root nodules of pulses help the plant by converting air's nitrogen into a usable form. This partnership is called:

(A) symbiosis
(B) digestion
(C) parasitism
(D) predation

38. A simple low-cost sanitation toilet that needs no flush water and turns waste into manure is the:

(A) open drain
(B) flush toilet on a sewer
(C) septic tank with soak pit
(D) compost toilet

39. One side of a square room is doubled to make a bigger square. The new area compared with the old is:

(A) 2 times as large
(B) 4 times as large
(C) 3 times as large
(D) 8 times as large

40. Standard (scientific) form writes 4500 as 4.5 x 10 raised to a power. That power is:

(A) 1
(B) 4
(C) 3
(D) 2

41. The part of a green plant that acts as the main 'food factory', where most photosynthesis happens, is the:

(A) flower
(B) root
(C) leaf
(D) seed

42. Leaving human waste in the open is dangerous because flies and contaminated water can spread diseases such as:

- **(A)** cholera and typhoid
- **(B)** malaria and dengue
- **(C)** asthma and allergy
- **(D)** scurvy and rickets

43. A path runs around the inside edge of a square plot. If the plot's perimeter is 48 m, each side of the square is:

- **(A)** 24 metres
- **(B)** 48 metres
- **(C)** 12 metres
- **(D)** 6 metres

44. The value of 3 raised to the power 4 is:

- **(A)** 81
- **(B)** 64
- **(C)** 27
- **(D)** 12

45. Cells around each stoma that swell and shrink to open or close the pore are called:

- **(A)** nerve cells
- **(B)** xylem cells
- **(C)** root hairs
- **(D)** guard cells

46. Open drains and uncovered waste are a serious health risk partly because stagnant dirty water becomes a breeding place for:

- **(A)** disease-carrying mosquitoes and flies
- **(B)** oxygen bubbles
- **(C)** useful honeybees
- **(D)** clean drinking water

47. A rectangular tile covers 48 cm^2 of surface and measures 8 cm along its length. Working backwards, its breadth must be:

- **(A)** 384 cm
- **(B)** 6 cm
- **(C)** 8 cm
- **(D)** 40 cm

48. Comparing 2■ and 5^2, the larger value is:

- **(A)** they are equal
- **(B)** neither, both are 10
- **(C)** 5 to the power 2
- **(D)** 2 to the power 5

49. Non-green plants and all animals ultimately depend on green plants for food, which is why green plants are called the:

- **(A)** parasites
- **(B)** consumers
- **(C)** decomposers
- **(D)** producers

50. The thick sludge collected from sewage is often put into a sealed tank without air, where bacteria digest it and release a useful fuel gas called:

- **(A)** petrol
- **(B)** oxygen
- **(C)** biogas
- **(D)** chlorine

51. To find how many square tiles of side 1 m cover a floor 5 m by 3 m, you should compute the floor's:

- **(A)** perimeter
- **(B)** height
- **(C)** diagonal
- **(D)** area

52. Expressed as a power of 10, the number one lakh (1,00,000) is:

- **(A)** 10 to the power 6
- **(B)** 10 to the power 5
- **(C)** 5 to the power 10
- **(D)** 10 to the power 4

53. To put back the nitrogen that crops remove from a field, a farmer can grow a crop of pulses such as gram or beans because their roots:

- **(A)** host nitrogen-fixing bacteria
- **(B)** turn the soil acidic
- **(C)** block water from the soil
- **(D)** burn nitrogen out of the soil

54. Citizens can help wastewater treatment work better by NOT throwing into the drain things like:

- **(A)** treated water from the plant
- **(B)** tea leaves, cotton, sanitary towels and plastic
- **(C)** dissolved oxygen
- **(D)** only clean rainwater

55. The area of a rectangle is 56 m^2 and its breadth is 7 m. Its perimeter is:

- **(A)** 15 metres
- **(B)** 28 metres
- **(C)** 56 metres
- **(D)** 30 metres

56. The prime factorisation 2 x 2 x 3 x 3 x 3 written using exponents is:

- **(A)** 2 cubed times 3 squared
- **(B)** 2 times 3 to the power 5
- **(C)** 6 to the power 5
- **(D)** 2 squared times 3 cubed

57. A lichen growing as a grey crust on a rock is actually a partnership of two organisms: a fungus together with a(n):

- **(A)** alga
- **(B)** bacterium
- **(C)** moss
- **(D)** insect

58. The final treatment step where the cleaned water is disinfected, often by adding chlorine, is meant to:

- **(A)** add nutrients for plants
- **(B)** kill any remaining harmful germs
- **(C)** make the water fizzy
- **(D)** turn the water into ice

59. A wire 36 cm long is bent into a square. The area enclosed by the square is:

- **(A)** 144 square cm
- **(B)** 36 square cm
- **(C)** 81 square cm
- **(D)** 9 square cm

60. Folding a sheet of paper in half doubles the layers each time. After 8 folds the number of layers is:

- **(A)** 2 to the power 8, which is 256
- **(B)** 8 to the power 2, which is 64
- **(C)** 2 to the power 7, which is 128
- **(D)** 2 to the power 8, which is 16

61. Carbon dioxide needed for photosynthesis enters the leaf chiefly through its:

- **(A)** stomata
- **(B)** flowers
- **(C)** waxy cuticle
- **(D)** roots

62. A household uses about 150 litres of water a day and sends roughly four-fifths of it down the drain as wastewater. The wastewater made each day is:

- **(A)** 30 litres
- **(B)** 120 litres
- **(C)** 150 litres
- **(D)** 75 litres

63. The distance around a circular flower bed is called its:

(A) circumference
(B) diameter
(C) area
(D) radius

64. A bacterium in an aeration tank splits into 2 every 20 minutes. In 2 hours (six splits) one bacterium becomes:

(A) 6 to the power 2, which is 36
(B) 2 to the power 6, which is 12
(C) 2 to the power 5, which is 32
(D) 2 to the power 6, which is 64

65. A destarched plant is kept in sunlight with one leaf partly covered by black paper. After the starch test, the covered part stays pale because that part:

(A) received no light and made no food
(B) was the oldest part of the leaf
(C) was given extra water
(D) had no chlorophyll

66. An aeration tank's floor is a rectangle 12 m long and 5 m wide. The area of its floor is:

(A) 34 square metres
(B) 17 square metres
(C) 60 square metres
(D) 120 square metres

67. A rectangular garden 20 m by 15 m has a 1 m wide path running all around it on the OUTSIDE is not asked; instead, the garden's own area is:

(A) 300 square metres
(B) 300 metres
(C) 35 square metres
(D) 70 square metres

68. A leaf cell divides into 2 each day. Starting from 1 cell on Monday, the number of cells on Friday (after 4 divisions) is:

(A) 2 to the power 4, which is 16
(B) 4 to the power 2, which is 16
(C) 2 to the power 3, which is 8
(D) 2 to the power 4, which is 8

69. Water and dissolved minerals taken up by the roots are carried up to the leaves through tube-like vessels called:

(A) cuticle
(B) phloem
(C) xylem
(D) stomata

70. Bacteria in an aeration tank double every hour. Starting from 1 colony, the number of colonies after 6 hours, written with a power, is:

- **(A)** 6 to the power 2, which is 36
- **(B)** 2 to the power 5, which is 32
- **(C)** 2 to the power 6, which is 12
- **(D)** 2 to the power 6, which is 64

71. Two identical right triangles, each of base 8 cm and height 5 cm, are joined to form a rectangle. The rectangle's area is:

- **(A)** 13 square cm
- **(B)** 40 square cm
- **(C)** 80 square cm
- **(D)** 20 square cm

72. A treatment plant cleans water in batches that grow ten-fold: 10 L, then 100 L, then 1000 L. The fourth such batch, as a power of ten, is:

- **(A)** 4 to the power 10 litres
- **(B)** 40 litres
- **(C)** 10 to the power 4 litres
- **(D)** 10 to the power 3 litres

73. The simple sugar first made during photosynthesis, before it is stored as starch, is:

- **(A)** protein
- **(B)** oxygen
- **(C)** fat
- **(D)** glucose

74. A treatment plant cleans 1000 litres in the time it takes to clean 10 x 10 batches of 10 litres. Written as a power of ten, 1000 is:

- **(A)** 10 to the power 4
- **(B)** 100 to the power 3
- **(C)** 10 to the power 3
- **(D)** 10 to the power 2

75. A leaf shaped like a rectangle is 6 cm by 2 cm. A caterpillar eats a square hole of side 1 cm in it. The leaf area left is:

- **(A)** 1 square cm
- **(B)** 12 square cm
- **(C)** 11 square cm
- **(D)** 13 square cm

76. A square photosynthesis tray has 10 rows and 10 columns of cells. Written as a power, the total number of cells is:

- **(A)** 10 squared, which is 100
- **(B)** 10 squared, which is 20
- **(C)** 2 to the power 10, which is 1024
- **(D)** 10 cubed, which is 1000

77. A leaf has roughly the shape of a rectangle 8 cm long and 3 cm wide. Treating it so, the sunlit surface area of one side is about:

- **(A)** 22 square cm
- **(B)** 11 square cm
- **(C)** 24 square cm
- **(D)** 16 square cm

78. A square settling tank has each side 7 m long. The length of fencing needed to run once around its top edge (its perimeter) is:

- **(A)** 49 metres
- **(B)** 11 metres
- **(C)** 28 metres
- **(D)** 14 metres

79. A square settling tank covers 64 m^2. The length of railing to fit once around its edge is:

- **(A)** 64 metres
- **(B)** 16 metres
- **(C)** 32 metres
- **(D)** 8 metres

80. Pond algae double their covered area every week. If they cover 3 m^2 now, after 3 weeks they cover:

- **(A)** 24 square metres
- **(B)** 12 square metres
- **(C)** 6 square metres
- **(D)** 9 square metres

81. One leaf makes about 5 mg of starch in an hour. A branch of 12 such leaves, working for 3 hours, makes a total starch of:

- **(A)** 60 mg
- **(B)** 36 mg
- **(C)** 180 mg
- **(D)** 20 mg

82. A town produces 2400 litres of sludge a day. If a digester can be filled by collecting equal amounts over 8 hours, the sludge gathered each hour is:

- **(A)** 19200 litres
- **(B)** 8 litres
- **(C)** 240 litres
- **(D)** 300 litres

83. A photosynthesis tray holds plants in a grid 4 rows by 5 columns of 1 cm squares. The total lit area of the grid is:

- **(A)** 9 square cm
- **(B)** 20 square cm
- **(C)** 18 square cm
- **(D)** 45 square cm

84. A sample shows the bacteria count rising as 3, 3^2, 3^3 each hour. After 4 hours the count is 3■, which equals:

- **(A)** 81
- **(B)** 12
- **(C)** 27
- **(D)** 64

85. A single mesophyll cell divides so the number of food-making cells doubles each round: 1, 2, 4, 8, … After 5 rounds the count of cells, written as a power of 2, is:

- **(A)** 2 to the power 4, which is 16
- **(B)** 5 to the power 2, which is 25
- **(C)** 2 to the power 5, which is 32
- **(D)** 2 to the power 5, which is 10

86. Out of every 100 litres of treated water, a park reuses 35 litres for watering. The percentage of treated water reused for the park is:

- **(A)** 35 percent
- **(B)** 65 percent
- **(C)** 100 percent
- **(D)** 3.5 percent

87. A rectangular sewage screen is 3 m wide and has area 21 m^2. Its length is:

- **(A)** 63 metres
- **(B)** 3 metres
- **(C)** 7 metres
- **(D)** 18 metres

88. The thickness of folded paper doubles each fold. Comparing 3 folds with 5 folds, the 5-fold stack is thicker by a factor of:

- **(A)** 8
- **(B)** 32
- **(C)** 4
- **(D)** 2

89. A square patch of clover used to fix nitrogen measures 9 m along each side. The area of soil it enriches is:

- **(A)** 90 square metres
- **(B)** 81 square metres
- **(C)** 36 square metres
- **(D)** 18 square metres

90. A rectangular grit tank is 10 m long and 4 m wide. The distance you walk going once around its edge is:

- **(A)** 20 metres
- **(B)** 14 metres
- **(C)** 40 square metres
- **(D)** 28 metres

91. A square plot's side is increased from 5 m to 10 m. The increase in its area is:

- **(A)** 50 square metres
- **(B)** 75 square metres
- **(C)** 100 square metres
- **(D)** 25 square metres

92. One litre of treated water is split among 10 test tubes, then each tenth is split among 10 more. The number of tiny final samples is:

- **(A)** 1000, which is 10 cubed
- **(B)** 100, which is 10 squared
- **(C)** 20, which is 10 plus 10
- **(D)** 10, which is 10 to the power 1

93. A plant packs the energy of about 10 x 10 x 10 sunlight-units into one starch grain. Written as a power of ten, that energy is:

- **(A)** 10 to the power 2
- **(B)** 30
- **(C)** 10 to the power 3
- **(D)** 3 to the power 10

94. A plant treats 5 lakh litres of sewage a day. Written in the short power-of-ten form (5 x 10 raised to a power), 500000 is:

- **(A)** 5 times 10 to the power 4
- **(B)** 50 times 10 to the power 5
- **(C)** 5 times 10 to the power 6
- **(D)** 5 times 10 to the power 5

95. A rectangular field 30 m by 20 m is to be watered with treated wastewater at 2 litres per square metre. The water needed is:

- **(A)** 100 litres
- **(B)** 600 litres
- **(C)** 1200 litres
- **(D)** 2400 litres

96. A field is watered in plots arranged 5 across and 5 down, each plot taking 2^2 litres. Written with exponents, the total water used is:

- **(A)** 2 to the power 5, which is 32 litres
- **(B)** 5 squared times 2 squared, which is 100 litres
- **(C)** 10 squared minus 1, which is 99 litres
- **(D)** 5 plus 2, which is 7 litres

97. A potted plant kept in a totally dark cupboard for many days grows pale and weak chiefly because, without light, its leaves cannot:

- **(A)** grow new roots underground
- **(B)** carry out photosynthesis to make food
- **(C)** breathe out carbon dioxide
- **(D)** absorb water through their roots

98. The main reason a town builds a wastewater treatment plant rather than letting sewage flow straight into a river is to:

- **(A)** collect rainwater for drinking
- **(B)** stop disease and protect the river's plants and animals
- **(C)** make the river flow faster
- **(D)** turn the river water sweet

99. A rectangular plot is 18 m long and 12 m wide. The cost of fencing it at 5 rupees per metre is:

- **(A)** 1080 rupees
- **(B)** 300 rupees
- **(C)** 150 rupees
- **(D)** 60 rupees

100. The value of $2^3 \times 5^3$, using the rule that equal exponents let you multiply the bases first, is:

- **(A)** 10 squared, which is 100
- **(B)** 7 cubed, which is 343
- **(C)** 10 cubed, which is 1000
- **(D)** 10 to the power 6, which is 1000000